

What LLM Trading Agents Actually Do in Production: A Six-Month, Population-Scale Record from Two Fleets

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Abstract

We present a continuous, population-scale measurement record of autonomous language-model trading agents operating under production conditions across two systems that share one design lineage: DX Terminal Pro (3,505 user-funded vaults trading real ETH in Base memecoin markets for 21 days, February to March 2026) and the DXAP live alpha fleet (500 to 599 user-created agents all-history, 91 to 117 concurrently active, trading Hyperliquid perpetuals, June to August 2026). The record spans roughly six months, 7.5M single-model invocations with about 300K onchain actions, and a further 231,638 multi-tool turns producing 14,596 fills. Four findings carry the paper. First, the operating layer (slider constraints, rendered candidate lists, order-path mechanics) determines behavior more than anything written in strategy text: a risk slider explains leverage (+0.425 per level), agent fixed effects absorb 60% of variance, and a leaderboard render boundary causally routes selection (regression discontinuity $1.75\times$ at the top-3 cut). Second, sizing is volatility-blind: median leverage is $5.0\times$ in every volatility sextile, and one posture-slider cell (11% of the book) holds 62% of liquidations. Third, agents capture almost none of the upside they reach: 43.2% of positions saw at least +300 bps of favorable excursion within 24h, yet 49.3% of those closed with a negative trade return; a mechanical bracket recovers +39.0 bps per position. Fourth, neither fleet shows a directional edge. The DXAP fleet is not profitable and trails a matched Hyperliquid retail benchmark (41% vs. 50% roundtrip win rate). A paired-replay league of frontier models on 416 captured production scenarios finds decision quality statistically indistinguishable across models at this horizon, while choice stability differs sharply across model families, and we preview the development program this null implies. Every headline survives day-clustered inference, permutation nulls, and a common-fee restatement, and the paper closes with a 17-rule methodology canon bought with our own retractions.

1 Introduction

By mid-2026, autonomous LLM agents routinely hold and move real money in public markets. Tournaments, arenas, and live platforms now put fleets of prompted models in front of crypto order flow, and a fast-growing literature evaluates them, largely by leaderboard P&L over horizons of hours to days. What the field lacks is a *record*: a sustained, population-scale account of what thousands of such agents actually do when users hand them configuration surfaces, strategy text, and capital, and of which parts of the surrounding system determine that behavior.

This paper assembles that record from two production systems built by the same laboratory on one design lineage. DX Terminal Pro (February to March 2026) ran 3,505 user-funded vaults, each a Qwen3-235B agent trading real ETH in a 12-token memecoin market on Base, for 21 days and 7.5M invocations.

The DXAP live alpha (June to August 2026) runs user-created agents on Hyperliquid perpetuals, mostly paper accounts at \$10,000 with live prices plus a small real-capital book, through a ten-tool turn loop. Our prior work [1] introduced DX Terminal Pro’s operating-layer architecture and failure-mode measurements, and explicitly deferred “deeper analysis of the live user-to-agent-to-execution traces.” This paper answers that deferral and extends it to the successor system.

Three commitments shape the presentation. (i) *The operating layer is the object of study.* We treat sliders, prompt templates, rendered market surfaces, and order-path mechanics as the experimental variables, because at production scale they are what actually varies. (ii) *Evidence classes are explicit.* Every headline claim is marked FIRM (registered analysis, day-clustered uncertainty, survives ablation) or PROVISIONAL (narrower or qualified); three of our own earlier results were retracted and are cited here only as retractions. (iii) *No performance claims.* Neither population shows a directional edge; the DXAP fleet loses money and trails its retail benchmark. We report this plainly, because the scientific value of a population-scale record is exactly its ability to establish such facts and to localize the defects that produce them.

Section 2 describes the two systems, the data stores, and the normalization and evidence-class rules. Section 3 gives the population and behavior overview against a matched retail benchmark. Section 4 shows the operating layer determines behavior; Section 5 covers risk behavior; Section 6 capture and exits; Section 7 the null results. Section 8 previews the development program these results motivate. Section 9 distills the methodology canon, and Sections 10–11 cover related work and limitations.

2 Systems and Data

2.1 Two systems, one design lineage

Table 1 summarizes the two deployments. The shared lineage is concrete: both systems expose the *same five-slider configuration* (trade activity/frequency, risk tolerance, trade size, holding style, diversification; each 1–5) plus free-text strategy with priority and expiry; both run a *one-action-per-turn loop* in which the model receives a compiled context and must emit typed tool calls with rationale; and both compile prompts from *Go templates* whose version lineage runs continuously from the DX Terminal Pro v2.x series into the DXAP Base/OPAL/JADE/Keystone families.

The differences are as designed: DX Terminal Pro was a bounded, real-capital tournament with a deliberately adversarial fee and a reaping mechanic that periodically eliminated the weakest token; DXAP is an open-ended alpha platform whose agents hold isolated-margin leveraged positions on a production perp venue.

2.2 Data stores and join discipline

DX Terminal Pro telemetry lives in Athena (`inference_log_payloads`) keyed by vault address / NFT id / request id, reconciled against onchain vault state; the public Dune dashboard accompanies [1]. DXAP telemetry lives in a ClickHouse analytics store: paper fills with realized P&L, fees, leverage, and liquidation price per fill; per-turn inference logs; valuation snapshots at ~2.5-minute cadence; trigger, memory, and config-revision tables; a normalized paper/live fills projection; and 5,035 real-money fills that we flag throughout and never pool silently with paper fills. Historical attribution joins on turn id + attempt id + the template and config revision *resolved at that turn*, never the current roster template.

Table 1: The two production systems. DX Terminal Pro figures are the canonical run facts of [1] and its lab record; DXAP figures are from the live ClickHouse analytics store as of Aug 15, 2026.

	DX Terminal Pro	DXAP live alpha
Window	Feb 26 to Mar 18, 2026 (21 days)	Jun 8 to Aug 15, 2026 (69 days and running)
Venue	12 memecoin tokens, Uniswap V4 pools on Base	~99–100 perpetuals on Hyperliquid (incl. HIP-3 synthetics)
Capital	Real ETH: 4,984 ETH deposited, 4,589 withdrawn (92%)	Mostly paper at \$10,000 start with live prices; small real-capital book (5,035 real-money fills, 6 of 19 users, notionals \$14–\$2,000)
Population	3,505 funded vaults (3,454 active)	500–599 agents all-history; 91–117 concurrently active
Model	Qwen3-235B-A22B-Thinking-2507 via SGLang, temp 0.6	OpenRouter mix; qwen3.7-plus dominates (86/91 active on Jul 22)
Cadence	~5 min/agent; exactly one tool call (buy/sell/observe)	15m/30m/1h/4h schedules + agent-created triggers; 10-tool manifest, mandatory <code>finalize_turn</code>
Volume	7.5M invocations, ~300K onchain actions, ~\$20M, ~70B inference tokens	231,638 finalized turns; 14,596 fills
Fees	2.3%/swap (2.0% protocol + 0.3% LP)	Paper engine fee eras: 4.5 → 14.5 (Jun 22 22:00 UTC) → 5.5 bps/side

2.3 Normalization, pooling, and fee restatement

The two systems are related by design lineage rather than schema compatibility, so we pool at the *metric* level (slider-to-behavior mappings, trade rate, hold time, win rate, fee burden) rather than at the row level. By study design, the DXAP analysis treats the fleet as one population and does not segregate paper from live accounts; the real-money book is flagged wherever it overlaps an analysis, and the paper engine’s mechanics are disclosed in Section 11 and restated here: fills at live mark with *zero slippage*, *zero funding* paid across all 14,596 fills, and a 0.4% maintenance-margin placeholder that liquidates positions roughly $2\times$ later than real Hyperliquid margining would. Because the builder fee moved across eras (4.5 → 14.5 → 5.5 bps/side), every cross-era P&L figure is restated at a common 5.5 bps rate (canon rule 11, Section 9). At that common rate the fleet’s cumulative realized P&L over the Jun 8 to Jul 26 window is $-\$217\text{K}$, against $-\$148\text{K}$ at zero fee; fees are not the explanation for the fleet’s losses.

2.4 Evidence-class discipline

Claims carry one of four classes. FIRM: registered analysis on the full window, day-clustered uncertainty intervals, and at least one ablation or permutation check. PROVISIONAL: positive but narrow, qualified, or sensitive to a design choice we could not fully close. WEAK/RETRACTED: reported only to document the failure mode. Three results in this program’s history were retracted, a HIP-3 lag edge (a timestamp artifact) and two “only net-positive posture/edition” claims (killed by calendar-footprint checks), and they appear here only as retractions. All headline numbers in Sections 3–6 are FIRM unless explicitly tagged.

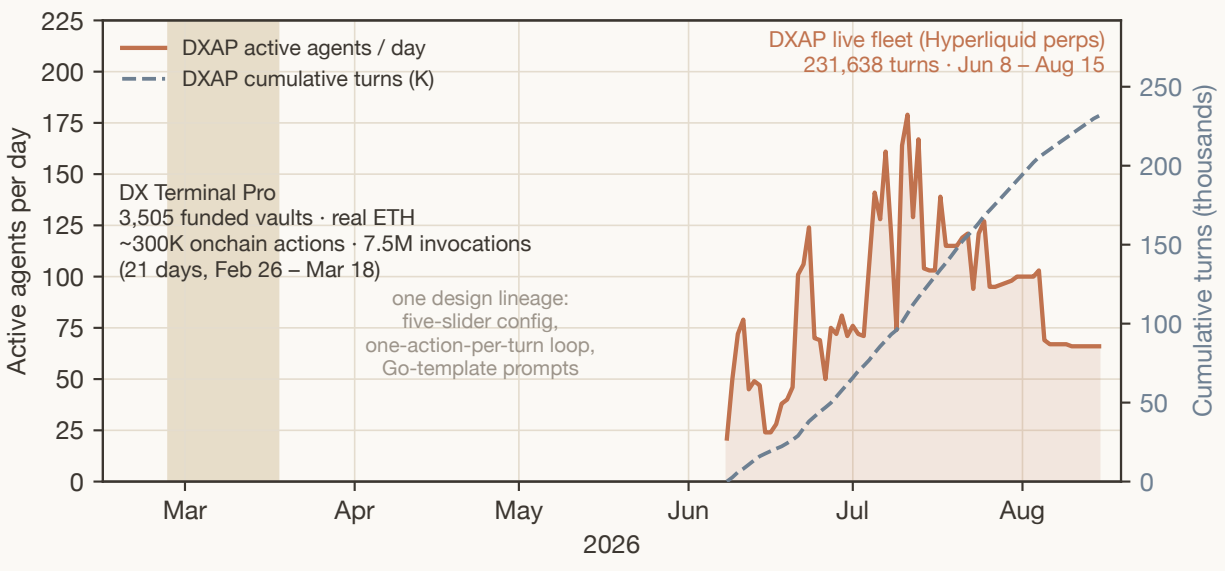


Figure 1: One continuous record across two production systems. The DX Terminal Pro window (shaded; Feb 26 to Mar 18) is drawn from the canonical run facts; DXAP daily active agents (solid, left axis) and cumulative finalized turns (dashed, right axis) are computed from the ClickHouse `analytics_agent_turns_finalized` table, Jun 8 to Aug 15, 2026.

3 Population and Behavior Overview

Figure 1 places the two deployments on one calendar. The record is continuous in the sense that matters for inference: one design lineage and one measurement discipline, with a two-month gap between deployments but none in the lab’s telemetry practice.

3.1 DX Terminal Pro in one paragraph

The 21-day tournament produced 7.5M invocations and $\sim 300\text{K}$ onchain actions from 3,505 funded vaults (3,454 active in the measured window); we use 3,505 for the funded population and 3,454 for activity-denominator statistics, reconciling the two counts that circulate in earlier lab documents. Behavior was bursty and strongly social: 1,544 of 3,454 active vaults bought the same token (FEET) within one hour on Mar 1 (peak 441 buys/min at 19:04 UTC); of the 908 launch-day buyers, the 82 still holding at measurement were 89% underwater (mean P&L -20.1%). The system recorded 3,878 sell cascades (≥ 10 vaults per 10 min), and 92.9% of trades occurred in two-sided 5-minute windows, so flow was contested in both directions. Portfolios converged over the run (mean pairwise Jaccard $0.304 \rightarrow 0.473$). Median true return was $0.492\times$; 16.2% of vaults finished profitable; 92% of deposited ETH was withdrawn.

3.2 The DXAP fleet against a matched retail benchmark

The daily measurement pipeline compares DXAP agents with Hyperliquid retail traders over matched windows (leaderboard sample of 1,961 traders, 1,000–100,000 equity band). As of Aug 15, over the trailing 4-day window (Table 2, Figure 2): the fleet runs a 41% roundtrip win rate against retail’s 50%, at nearly identical median hold times (2.05h vs. 2.18h); only 15% of agents active over the week are net-positive versus 53% of retail accounts; agents lean structurally long ($\sim 78\%$ of entries vs. $\sim 60\%$) at roughly $4\text{--}5\times$ the effective leverage (median $0.096\times$ vs. $0.022\times$ notional-to-equity, and a chosen-leverage median of $5.0\times$).

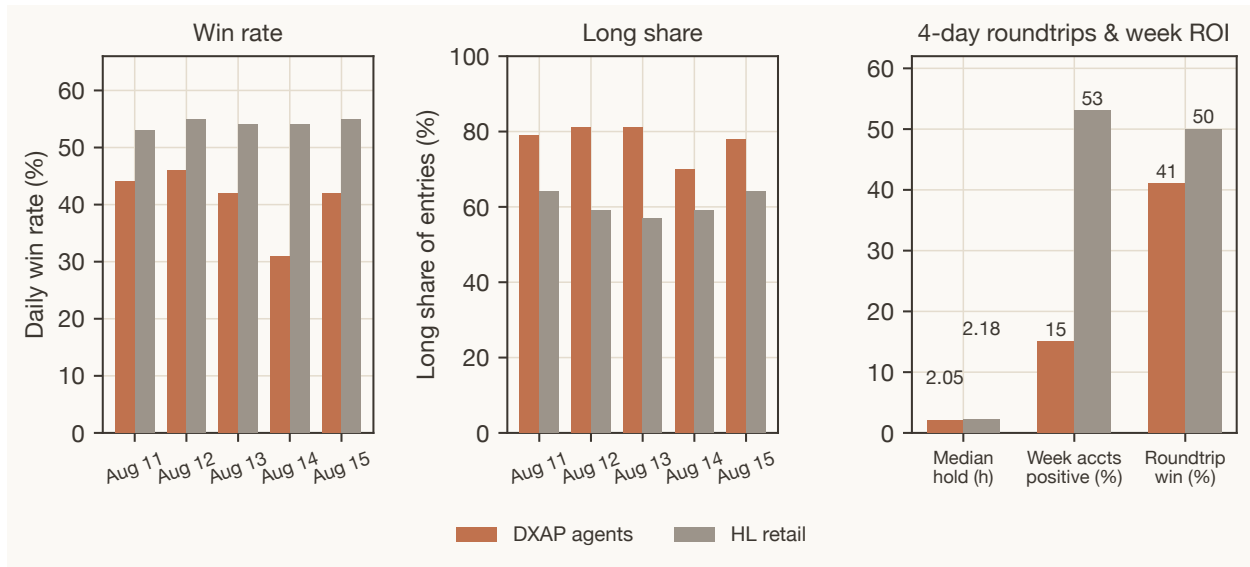


Figure 2: DXAP agents vs. a matched Hyperliquid retail sample, trailing window ending Aug 15, 2026. Left: daily win rate. Center: long share of entries (count-based). Right: 4-day open-to-close roundtrips (ours $n = 353$, retail $n = 16,009$) and share of week-active accounts with positive net ROI. Source: `h14d_summary.json`, daily pipeline of Aug 15, 14:00 UTC.

Table 2: Fleet vs. retail benchmark, trailing windows ending Aug 15, 2026. Retail sample: 1,961 leaderboard traders in the \$1K–\$100K equity band.

Metric (window)	DXAP agents	HL retail
Roundtrip win rate (4d)	41% ($n = 353$)	50% ($n = 16,009$)
Median hold (4d)	2.05 h	2.18 h
Median roundtrip return (4d)	−0.094%	+0.002%
Accounts net-positive (week)	15% ($n = 117$)	53% ($n = 4,828$)
Long share of entries (count)	~78%	~60%
Effective leverage, median	0.096×	0.022×
Chosen leverage, median	5.0×	—
Return by hold bucket	<1h: −0.15%; 24h+: +0.23%	—

The fleet is not profitable: cumulative realized P&L stands at −\$217K at the common 5.5 bps fee rate, and no user was net-positive at the Jul 12 census. Its one relative edge is horizon: positions held 24h+ earn +0.23% median account return while sub-1h positions earn −0.15%.

The two populations bracket the design space: DX Terminal Pro shows what a frozen single-model harness does at 3.5K-agent scale under real capital and a 2.3% fee; DXAP shows what an open user-configured fleet does over months against live prices. The remaining sections are organized by what the record establishes rather than by system.

4 The Operating Layer Determines Behavior

The strongest cross-system regularity in this record is that behavior is set by the machinery around the model (configuration surfaces, rendered candidate lists, order-path mechanics) more than by anything the model decides in text.

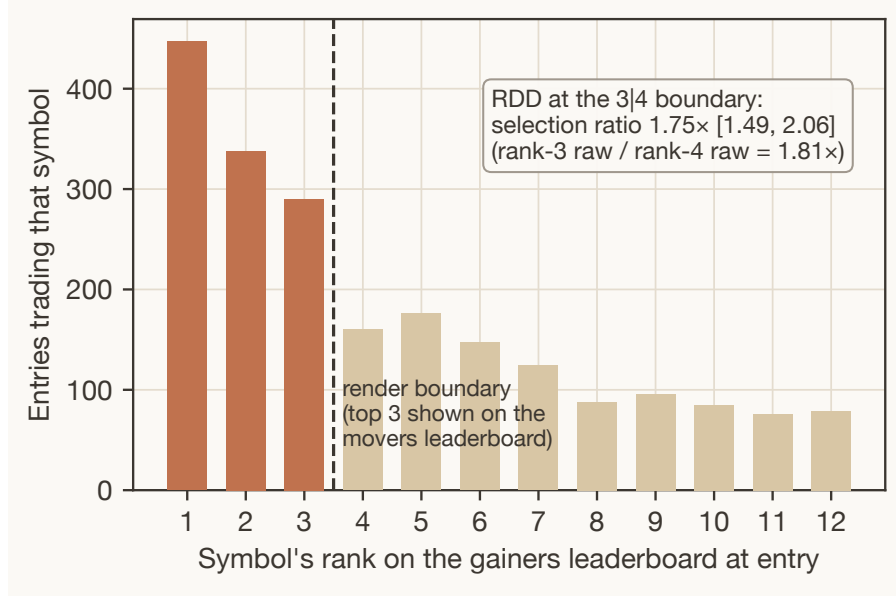


Figure 3: Selection by gainers-leaderboard rank at entry (Jun 16 to Jul 24; 2,339 entry observations across gainers ranks 1–15, plotted through rank 12). Terracotta bars are rendered ranks; the dashed line is the render boundary. The registered RDD estimate at the 3|4 cut is $1.75\times [1.49, 2.06]$; the raw rank-3/rank-4 count ratio is $1.81\times (290/160)$.

4.1 Sliders set leverage; identity explains the rest

In DXAP, chosen leverage is essentially a configuration constant. The `riskTolerance` slider maps to leverage at $+0.425\times$ per level ($p = 3.8 \times 10^{-279}$); agent fixed effects absorb 60% of variance; and when users name a leverage in strategy text, realized leverage tracks it at Spearman $\rho = +0.836$. Nothing else we measured (market state, recent P&L, prompt family) moves sizing comparably.

4.2 A render boundary causally routes selection

DXAP agents see a “movers” leaderboard rendering nine symbols: the top three gainers, losers, and volume leaders. 46.5% [43.4, 49.7] of entries are in rendered symbols against an 8.9% random-availability baseline, a $5.2\times$ over-selection that holds for every strategy posture and every prompt template. The render boundary itself is causal: a regression discontinuity across the rank-3/rank-4 cut gives a selection ratio of $1.75\times [1.49, 2.06]$ *exactly* on the boundary (Figure 3); symbols just below the cut are statistically identical in their market state but are picked far less because they are not shown. This is the one clean causal result in the record, and its cause is a rendering choice. The router is honest (re-routing costs \$0–17K depending on assumption) and decisive.

4.3 Strategy text vs. constraints: a cross-system contrast

The two systems resolve text–control conflicts in opposite directions. In DX Terminal Pro, sliders act as constraints that override strategy text: mandate fidelity follows slider settings (e.g. trade-activity gradients across slider levels), and agents at the lowest activity setting with insistent strategy text still trade at slider rates (3.95% vs. 1.51% invocation trade rate: elevated, but bounded by the slider). In DXAP, a user-behavior study found the reverse failure: strategy text routinely overrode the frequency slider, with agents trading far above configured cadence when the text demanded it. Same five-slider lineage, opposite conflict-resolution

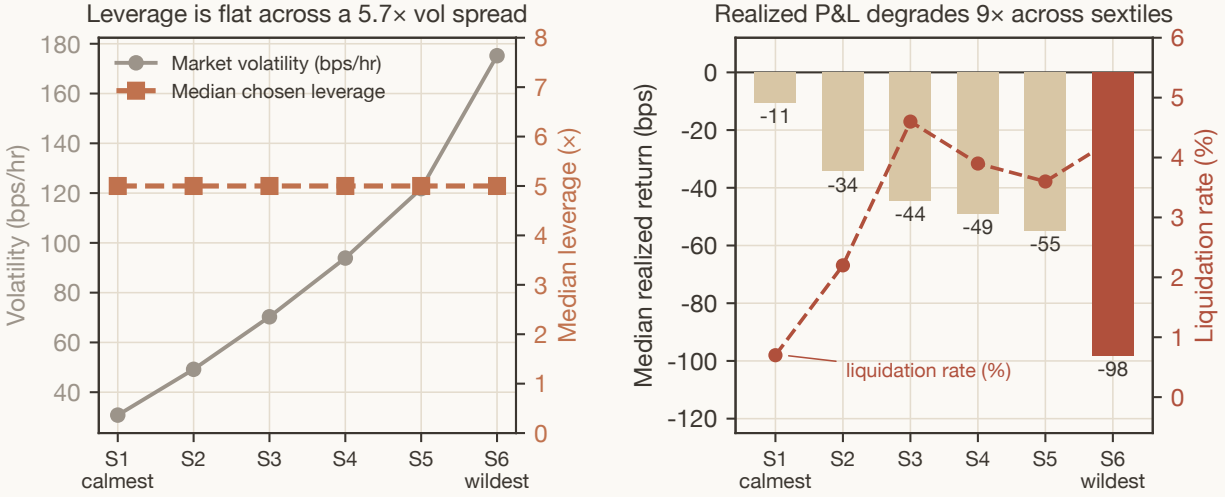


Figure 4: Volatility-blind sizing across 6,400 closed DXAP positions. Left: market volatility per sextile vs. median chosen leverage (flat at 5.0 $\times$). Right: median realized return per sextile (bars) and liquidation rate (dashed).

outcomes. Which surface wins is therefore an operating-layer decision, and it must be chosen deliberately, because users read the sliders as commitments. A related DX Terminal Pro finding sharpens the point: owners who wrote concrete, numeric instructions were profitable 4.2 $\times$ as often as the median, and the 87 UI-only owners who never used chat were the highest-profit cohort (41% in profit). Structured control surfaces outperformed conversation as an interface to agent behavior.

5 Risk Behavior

5.1 Sizing is volatility-blind

The largest single defect in the DXAP record is that sizing ignores volatility. Measuring each entry’s market volatility as the standard deviation of the prior 24 hourly returns and splitting all 6,400 closed positions (Jun 8 to Jul 24) into sextiles: median leverage is 5.0 $\times$ in *every* sextile across a 5.7 $\times$ volatility spread (Figure 4, Table 3); Spearman(volatility, leverage) = -0.001 ($p = 0.92$). Notional goes the wrong way: Spearman(volatility, notional) = $+0.165$ ($p = 2.7 \times 10^{-40}$), so agents *size up* in wilder names. Median realized return degrades from -10.6 bps in the calmest sextile to -98.2 bps in the wildest, and the liquidation rate rises from 0.7% to 4.3%. The gradient is not an artifact of liquidations: excluding all 205 of them, realized return still falls from -8.5 to -53.6 bps across sextiles ($\rho = -0.104$, $p = 1.8 \times 10^{-16}$, $n = 6,195$), while bracketed entry quality stays flat. Leverage and stop geometry are both volatility-blind.

5.2 Liquidation risk is concentrated in one posture-slider cell

Liquidations cluster in one cell of the book. Of 205 liquidations among 6,400 closed positions (3.2%, carrying 74% of gross loss), 128 (62%) sit in a single cell: momentum-posture agents at frequency-slider 5, roughly 11% of the book (Figure 5). A Mantel–Haenszel estimator stratified by day gives odds ratio 22.37 [12.59, 37.45]. The cell spans 70 distinct agents, and across 411 momentum-posture positions at frequency < 5 there were *zero* liquidations. Risk in this fleet is a configuration choice made at setup time rather than a series of bad in-context decisions.

Table 3: Closed positions by entry-time volatility sextile (Jun 8 to Jul 24). Realized return is median net bps; bracketed-entry quality is flat throughout, so the degradation is sizing, not selection.

Sextile	<i>n</i>	Vol (bps/hr)	Med. lev	Med. notional	Liq. rate	Med. realized
S1 calmest	1,067	30.8	5.0×	\$3,500	0.7%	−10.6
S2	1,067	49.2	5.0×	\$4,000	2.2%	−34.3
S3	1,066	70.3	5.0×	\$5,000	4.6%	−44.5
S4	1,067	93.9	5.0×	\$5,140	3.9%	−49.0
S5	1,066	121.8	5.0×	\$5,405	3.6%	−54.9
S6 wildest	1,067	175.2	5.0×	\$5,000	4.3%	−98.2

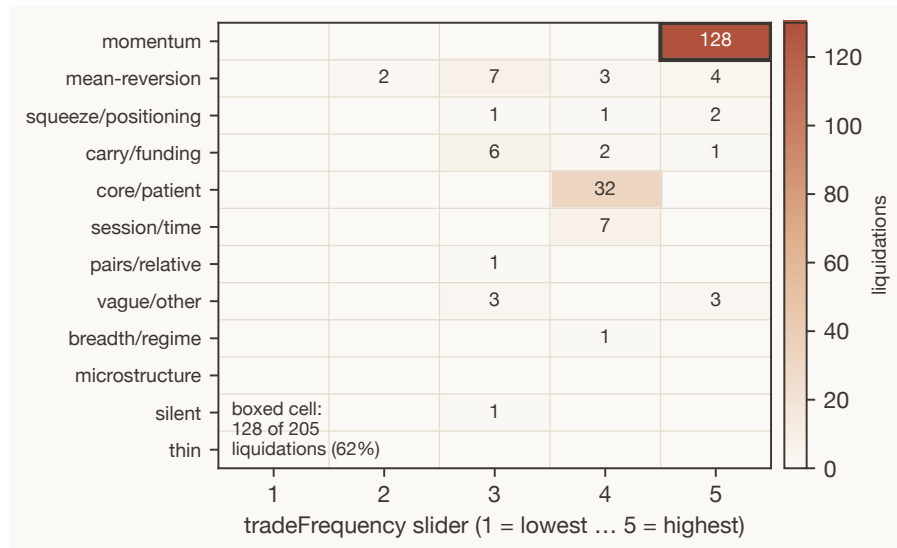


Figure 5: Liquidations by strategy posture × tradeFrequency slider (6,400 closed positions, 205 liquidations). The boxed cell (momentum × frequency 5) holds 128 of 205 (62%). Cell counts are from the style-lab position master; the odds ratio is the day-stratified Mantel–Haenszel estimate.

5.3 Forced numeric restatement fails

The strongest prompt-side lever the template program found was requiring the model to compute and state its liquidation distance before entry. It fails: agents state liquidation distance in 45.0% of entry turns and size identically anyway, and staters liquidate *more* often (5.8% vs. 1.2% for non-staters), because stating marks aggressive intent rather than restraining it. We read this as the most consequential negative result for prompt-based control in the record: when the dangerous parameter is set by a configuration constant, no amount of in-context arithmetic intervenes on it. The fix has to live at the order path (canon principle: mechanism over exhortation; see Section 9).

5.4 A preregistered probe of implicit risk representations (H2)

We preregistered (Jul 25 to 26) a narrow question: does a neural representation of the rendered pre-action state carry liquidation information beyond 25 concrete features? Target: liquidation (base rate 3.20%, 205/6,390); frozen concrete-feature baseline PR-AUC 0.1145, ROC 0.822. The result is narrowly positive and heavily qualified; we report it as PROVISIONAL:

- A small encoder reading the rendered turn card beats concrete features by +0.0150 PR-AUC, 90%

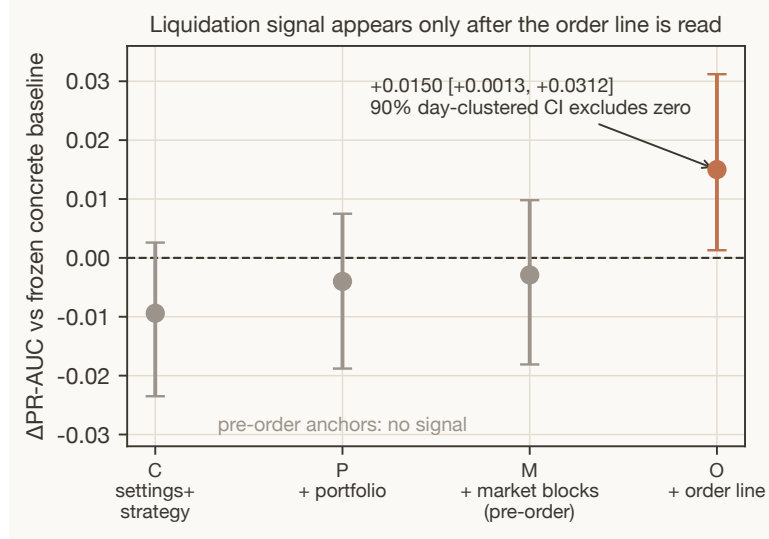


Figure 6: H2 anchor ladder: Δ PR-AUC over the frozen concrete baseline when pooling at successive points in the turn card. Error bars are 90% day-clustered intervals. The entire increment sits at the order boundary.

day-clustered CI $[+0.0013, +0.0312]$ excluding zero; a TF-IDF arm scores -0.0056 (chance), so the increment is not lexical. It survives adding the traded symbol’s rendered market row ($+0.0012$ extra, CI spans zero), the full rendered state undiminished ($+0.0153$), 78 pairwise interactions ($+0.0170$), and 10/10 PCA seeds.

- **The anchor ladder qualifies everything.** Pooling at four anchors in the same card (Figure 6): the increment appears *only after the order line is read*. Pre-order anchors add nothing (-0.0094 , -0.0040 , -0.0029 ; all CIs span zero). Post-order, side and symbol decode at 0.998 – 0.999 , so the representation is echoing the order text, and liquidation legibility jumps from 0.571 – 0.615 to 0.712 . Even the consecutive pre-order→post-order delta ($+0.0179$ $[-0.0026, +0.0430]$) does not itself clear zero.
- **Scope.** The trading decisions were made by qwen3.7-plus behind an API; the probe reads activations of Qwen3.5-4B reading the same card. The claim is therefore “a neural encoder of this state carries risk information,” never “the agent knew.”

The operational consequence: there is no pre-decision risk signal to harvest. The deployable form is a *post-order check*. At a 10% flagging budget the representation catches 71/205 liquidations vs. 57/205 for concrete features (precision 11.1% vs. 8.9%), which makes it useful for screening orders already written rather than for warning about situations.

6 Capture and Exits

6.1 The capture gap

The fleet’s positions frequently go somewhere profitable; the fleet rarely keeps it (Figure 7). 43.2% of closed positions (2,765 of 6,400) reached $\geq +300$ bps of maximum favorable excursion within 24h. Of those, 49.3% closed with a negative trade return ($\text{ret. bps} < 0$); only 13.3% kept half the excursion. Book-wide median favorable excursion is $+247.9$ bps against a median realized -33.3 bps; median capture where upside existed is 2.0%. Upside is nearly as predictable as risk from pre-trade state (ROC 0.751 vs. 0.783 for liquidation), yet capture is not (ROC 0.541, essentially chance): the market state predicts what the position

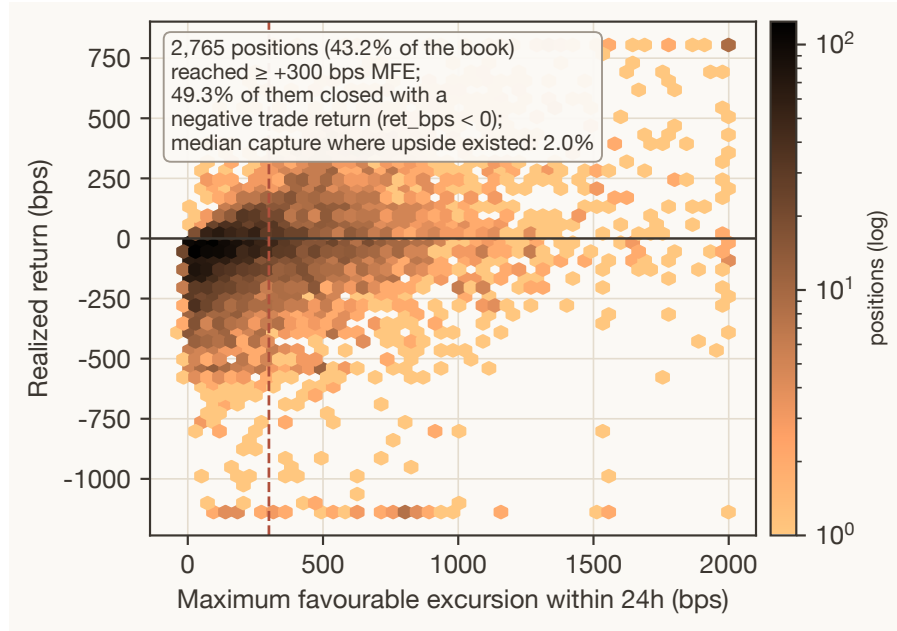


Figure 7: The capture gap: maximum favorable excursion within 24h vs. realized return for 6,400 closed positions (hexbin, log counts; clipped at the 0.5/99.5 percentiles for display). The dashed line marks +300 bps MFE.

will *do*, and it does not predict what the agent will *keep*. Selecting for predicted upside currently makes P&L *worse* (the top predicted-run quintile sees median MFE of +410 bps and median realized -71.0 bps), and widening stops in the wildest quartile is harmful (-68.9 [$-115.5, -17.0$] bps).

6.2 Mechanical exits beat discretionary exits

The best-measured behavioral lever in the record is an order-path mechanic: attaching a fixed 2%/4% stop/target bracket at entry. Paired, day-clustered, the bracket earns +39.0 bps per position [$+21.3, +56.5$]; excluding all liquidations it still earns +16.6 [$+2.1, +30.8$]. Roughly 60% of the gain is blow-up prevention (+717 bps across the 205 liquidations); the remainder is improved giveback on ordinary positions. 11 of 16 exit policies clear zero; none is profitable outright (best -9.5 [$-25.1, +8.2$]). Exit discipline lives in the tool surface: $\sim 83\%$ of entries in both arms state stop and target, 131 exits show zero plan-abandonment, and agent journals sit empty 10/10. The DX Terminal Pro side shows the same pattern at crowd scale: POOPCOIN’s exit cascade (438 sells, median 9.5s gaps) is the same discretionary-exit failure, and its concrete-instruction result ($4.2\times$) points the same way. The more behavior is moved into explicit, checkable structure, the better it goes.

6.3 Why the bracket works: it truncates the left tail

The bracket’s effect sizes are exact about the mechanism: the majority of the gain comes from truncating the left tail. This matters for interpretation. The lever is a *risk mechanic*, and it preserves aggressive exposure rather than restricting it. Its current blocker is also mechanical: a non-retryable trigger quota error left 24 of 35 successful opens without their protective trigger in one 48-hour window. The highest-value change in the whole record is an atomic open-with-protection order path.

7 Null Results

A population-scale record earns its keep on its nulls. We list the major ones; all are FIRM unless noted.

No directional edge from any information source. A 77K-candidate signal ledger sorts at chance (leave-one-out win 0.50). White-box probes over 689 traces add +0.001–0.003 over concrete features, at or below the permutation null. Stuffing 770K tokens of context into a paired top-pick comparison yields −0.46 points [−2.93, +1.75]. A world-context arm went 0/84. The `market_research` sub-agent’s recommendations are null ($n = 120$). Agent selection is at chance: horizon-1 to horizon-2 rank correlation +0.018 ($p = 0.93$); block test $\rho = 0.0697$ ($p = 0.118$).

Nothing is day-clustered positive. Across 8 strategy postures, 6 prompt templates, and 9 cohorts, no group’s lower day-clustered confidence bound exceeds zero; the nearest (one post-launch cohort, +22.6 [−0.2, +47.0] bps) does not clear. This census is what forced the retractions of two earlier “only net-positive” claims (Section 2).

Chase-state over-selection without payoff. Entries in $> +0.75\%/1h$ states occur at $2.46\times$ availability; trailing-4h return at entry is +136.5 bps [+120.6, +151.9] while forward-4h is −6.9 [−15.8, +2.2]; venue short-horizon reversal runs $\rho = -0.0252$ per day, negative on 36 of 47 days; within-chase picks underperform random by −7.82 [−14.6, −1.2] bps at 1h. Corroborated on 4,096 DX Terminal Pro terminal entries (−3.774 bps per percentage point of prior-4h side-aligned return, two-way clustered): the only finding in this section that crosses systems, and it crosses as a null.

Memory is a contamination channel. Memory-write frequency correlates *negatively* with P&L ($\rho = -0.200$, $p = 0.002$). Ghost strategies occur: one agent ran 31 of 32 positions under strategy text that had been deleted, citing it three weeks later. Agents fabricate funding income on a zero-funding venue. 66% of 77,269 memory operations are compactions, and 40% of compactions *grow* the file. DX Terminal Pro’s production fabrication trajectory (13.3% $\rightarrow$ 6.1% of trades over three weeks; 41–54% of observations persistently) is the same phenomenon in a different store.

Restraint beats added information (provisional). The strongest positive template finding is negative in form: a qualitative-restraint arm scores +0.52 points [+0.16, +0.97] with the lowest intervention rate in its league, and intervention rate correlates with score at −0.72. We flag it PROVISIONAL because template league effects are exactly the claims this program has had to retract before; it awaits replication on a fresh window.

8 Toward an Edge: A Development Program

Everything above is backward-looking: it localizes where two production fleets lose money and why. We close the empirical part of the paper with the question those results raise. If the operating layer determines behavior, and no measured group shows a directional edge, where does a builder invest next? This section is the lab’s working answer, stated as a development program rather than a result. Each pillar is tied to a specific measurement above, and the ordering of the pillars is itself an evidence statement. Two pieces of forward-looking evidence appear here: a model-comparison league, which is *replay* evidence with the limitations that implies, and one internal training signal, labeled internal and early. Neither is a performance claim.

8.1 Harness engineering continues to pay first

Every measured positive lever in this record operates on the machinery around the model. The order-path bracket earns +39.0 bps per position [+21.3, +56.5] (Section 6); the render boundary routes selection $1.75\times$

at the top-3 cut, the one clean causal result in the record (Section 4.2); and the bracket’s own blocker was mechanical, a non-retryable trigger-quota error that left 24 of 35 successful opens unprotected in one 48-hour window. The program’s first pillar follows directly: an atomic open-with-protection order path, leverage scaled to realized volatility at the tool rather than requested in text, and deliberate A/B of the render. None of these requires a better model. All are cheap. And the failure modes they address (volatility-blind sizing, render-driven chase, uncaptured excursion) are where the record says the money goes.

8.2 Subagents and tool surfaces with demonstrated utility

The second pillar is a utility bar: a tool surface stays in the manifest only if it demonstrably changes behavior, and subagents count as tool surfaces. The structured-exits surface passes the bar: 131 bracketed exits show zero plan-abandonment, while agent journals sit empty in 10 of 10 sampled agents, so the exit discipline that exists lives in the tool (Section 6). The `market_research` sub-agent fails the informational version of the bar today: its recommendations are null at $n = 120$ (Section 7). The lab therefore now treats it as a *behavioral* surface rather than an informational one. Its test question becomes “does it measurably change what the agent does,” and any surface that cannot show such a change against a null arm within a registered window is removed. Measured restraint beats unmeasured capability.

8.3 Risk alignment as a deployable check

The H2 probe (PROVISIONAL, Section 5) found no harvestable pre-decision risk signal: the representation increment appears only after the order line is read. The deployable consequence is a *post-order representation check* that screens orders already written: at a constant 10% flagging budget it catches 71 of 205 liquidations against 57 of 205 for concrete features, a +25% liquidation catch at a fixed budget. The check’s companion is a decision-provenance ledger: every order carries the rendered state, the template revision, and the config revision resolved at that turn, which is the join discipline of Section 2 turned into a write path. The ledger is what makes the check auditable, and it is what makes the replay league of the next subsection possible at all. Both are infrastructure changes; neither touches the model.

8.4 Model training and environment building

Figure 8 presents the June 2026 model league: a paired replay of 416 captured production scenarios across 7 days at temperature 0.6, scored as winsorized regret against the achievable envelope, with day→8h-block hierarchical bootstrap intervals [3]. Three frontier models span 263.27–264.37 bps of regret. Every interval overlaps every other, and Holm-adjusted tests against the leader are all non-significant (minimum $p = 0.46$): no measurable decision-quality separation at this horizon. The forced-action arm (flat removed from the menu; 280 cells) sharpens the picture. Repeating the same scenario three times at temperature 0.6, claude-fable-5 changed its chosen (symbol, side) action in 35% of cells, against ~90–95% for the qwen3.7 pair: model families differ sharply in how repeatable their choices are. Two properties of this variation matter for interpretation. First, the strategies being executed are qualitative, so more than one action can satisfy a strategy’s intent in a given state; a changed choice is not automatically a wrong one. Second, the variation carried no measurable economic cost in this replay: selection quality across repeats spans rank percentile 0.486–0.508 (0.5 is chance), and regret intervals for all three models coincide. Within the tested envelope, the choices that vary across repeats performed the same as the ones that did not. Consistency and correctness are separate axes, and only the first separates these models. Economics separate them too: at tied decision quality, qwen3.7-plus completed the 416-cell league for \$8.25 in inference spend against \$203.61 for claude-fable-5, a roughly 25-fold cost difference that compounds at production turn volumes, which is one reason the production fleet runs the qwen3.7 pair.

The required caveat: this is replay evidence on captured scenarios, meaning regret against the achievable envelope under a frozen context, and it says nothing about live P&L. Per the lab’s evidence rules, replay can show how an agent behaves under a frozen context; it cannot promote a profitability claim, and we make none. What it can do, and does, is bound how much decision quality differs across models: at this horizon, by less than the replay noise floor.

The implication directs the fourth pillar. If model shopping buys nothing measurable, differentiation must come from training on the environment, and the continuous record itself *is* the environment: 231,638 finalized turns with resolved context, tool calls, and reconstructed fills are a turn-level training distribution that no public benchmark has. An early internal signal points this way: DX-SFT-0.3, a supervised fine-tune on 41,203 harness turns, improves on the lab’s internal DXTradeBench decision evaluation from 119 (baseline) to 96 (lower is better; internal lab report, early, not externally validated) [2]. The next step is live-branch GRPO: reinforcement on branched continuations of the production turn stream, with the canon of Section 9 as the reward-audit layer. We report the SFT number only to document direction; it claims no edge.

8.5 Benchmarking as infrastructure

The final pillar underwrites the other four. The model league is now recurring infrastructure, re-run on every model release; DXTradeBench versions the training signal; and the public versioned datasets at <https://dxrg.ai/research> keep both honest from the outside. This program’s own history is the argument for treating benchmarks as infrastructure: three of its claims were retracted, and one rolling p -value ran $0.0067 \rightarrow 0.19 \rightarrow 0.0277$ before anyone called it (Section 9). A benchmark you must report against is the scoreboard that keeps a development program from grading its own homework.

The ordering of these pillars follows the record’s effect-size map. Harness levers come first because they are FIRM, large, and cheap. Risk alignment comes second because it is PROVISIONAL but acts at the same order path and is deployable now. Training comes third because the league says the off-the-shelf decision cores are indistinguishable, so the only lever left on the model side is one trained on this environment. Benchmarking runs throughout because this lab has measured what happens without it. If a real edge ever arrives, this record says it arrives in that order.

9 A Methodology Canon for Agent-Trading Research

Every rule below was bought with a retraction or a failed claim in this program. We consider it the most transferable output of the record, and we state the rules as we now enforce them.

1. **The market day is the inferential unit, never the trade.** Positions opened the same day share the tape; position-level intervals here run $\sim 2.5\times$ too narrow.
2. **Pre-register the band, or report the entire sweep.** One of 35 nested definitions clearing zero is what chance looks like.
3. **Require an increment over a concrete baseline.** This killed three of our own “edges”; all were re-encodings of already-visible features.
4. **Leave-window-out cross-validation, never random splits.** Time-series data leaks across random folds.
5. **Permutation nulls on every arm.**

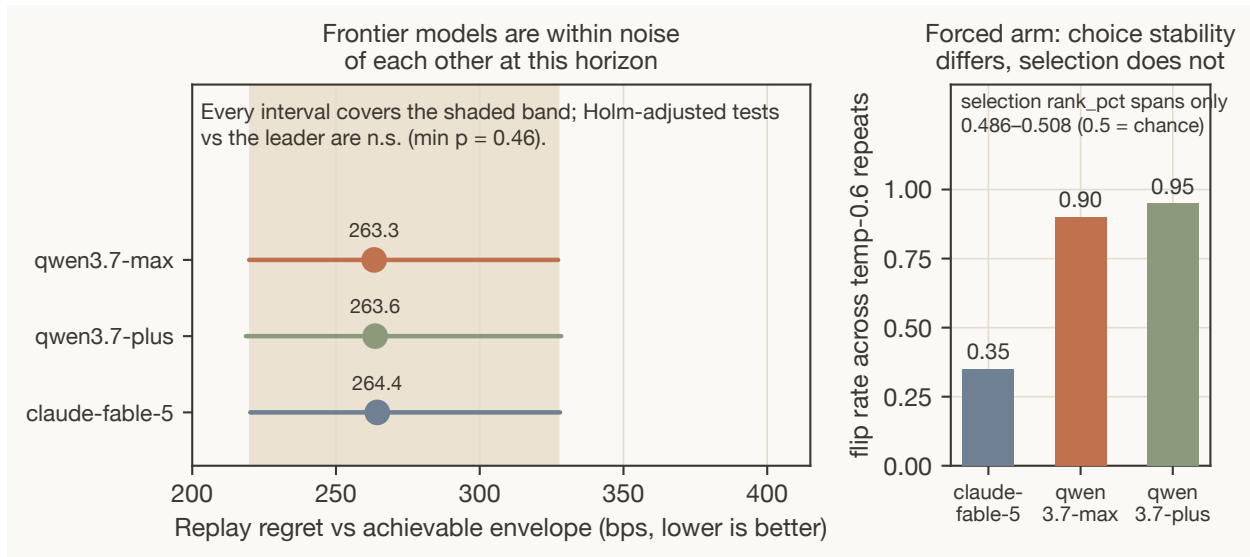


Figure 8: The June 2026 model decision league. Left: winsorized replay regret against the achievable envelope (bps, lower is better) for three frontier models over 416 captured production scenarios across 7 days at temperature 0.6; whiskers are day→8h-block hierarchical bootstrap intervals, and the shaded band is the range every interval covers. Right: choice stability in the forced-action arm (flat removed; 280 cells), the fraction of cells in which the model’s choice flips across temperature-0.6 repeats; selection quality spans rank percentile 0.486–0.508 (0.5 is chance). Replay measures behavior under a frozen context and says nothing about live P&L.

6. **Conditioning on trades taken is post-treatment, so it is descriptive only.** Removing the agent grew our state panel from 916 positions to 104,780 market-hours.
7. **Prefer revealed preference to outcomes.** Over-selection ratios carry no market-variance term and are readable in a day; outcome metrics need weeks.
8. **Agent text never promotes an experiment; reconstructed fills do.** Agent-reported P&L is non-canonical.
9. **Check the calendar footprint before believing any group difference.** This overturned two of our claims within an hour.
10. **Read the whole turn.** Bought with an argMax bug that read only the last message of five-message turns.
11. **Compare at a common fee rate.** Our builder fee moved 4.5 → 14.5 → 5.5 bps/side across eras.
12. **Check what the primary metric throws away.** Counting only closed positions censors half the lifecycles; if treatment changes hold time, the censoring is informative.
13. **Calibrate against a null arm before reading any P&L table.** Byte-identical templates produced ~\$3.5K of spread; anything smaller than your null arm’s spread is noise.
14. **Transport success is not tool success.** Error payloads ride inside successful envelopes; 14.9% of tool calls failed at result level in one league while transport reported success.
15. **A rolling p -value read many times is not a p -value.** One of ours ran 0.0067 → 0.19 → 0.0277.
16. **Beware habitat selection.** Retuning on early activation selects for “fired during this tape,” not “owns a durable mechanism,” and systematically penalizes rare-event postures.

17. **Know your timing-luck floor.** Identical contracts at a ± 15 -minute schedule offset produced -88 vs. $+21$, $+17$ vs. -38 , $+9$ vs. -75 dollars. Most effect sizes being compared are smaller.

We add one working principle that the canon implies: *mechanism over exhortation*. Every prompt-side attempt to fix a behavior that lives in the operating layer (forced restatement, added context, stronger wording) underperforms a one-line change to the order path or the render.

10 Related Work

Agentic trading evaluations. Public LLM-trading leaderboards, most visibly nof1’s Alpha Arena [4] (six models $\times$ \$10K real capital, run on Hyperliquid), pit frontier models against each other on live capital and have done more than any benchmark to popularize the domain; they are tournaments rather than measurement programs, and they publish P&L without operating-layer telemetry. KTD-Fin [8] evaluates model trading behavior with finer task decomposition, and the Agent Market Arena [5] studies multiple agents in a shared simulated market. TradingAgents [7] supplies the reference multi-agent role architecture much of this ecosystem builds on. Xia et al.’s survey [6] organizes the fast-growing literature on LLM agents in financial trading. None of these has access to the user-to-agent-to-execution trace at population scale, which is the gap this record fills.

Our own prior work. [1] introduced DX Terminal Pro’s architecture, its five measured failure modes, and frozen-harness production behavior, and deferred the trace-level analysis. Relative to it, the new material here is: the herd-return economics and cascade distributions; the production fabrication trajectory; the entire DXAP fleet record (volatility-blind sizing, render-driven selection, liquidation concentration, the capture gap, the bracket result, the H2 probe); the cross-system contrast in text–control resolution; and the methodology canon.

Benchmarks and harnesses. The lab’s MEMEBench and harness-transfer work [1] cover model-selection and EVM construction accuracy; this paper deliberately takes the harness as given and measures the population under it.

11 Limitations

Paper-engine mechanics. Most DXAP fills are paper: fills at live mark with zero slippage, zero funding paid across all 14,596 fills, and a 0.4% maintenance-margin placeholder that liquidates roughly $2\times$ later than real Hyperliquid margining. All cross-era P&L is restated at 5.5 bps/side, but restatement does not manufacture slippage or funding. The 5,035 real-money fills are flagged wherever relevant; per the study design they are analyzed as part of the one-fleet population with the flag visible rather than pooled silently or excluded. Liquidation-related results should be read as *earlier and larger* under real margining.

Thin positives. 205 liquidations underpin the tail results; the concentration finding is robust to stratification but the cell counts are small. **Single venue per era.** Base memecoins in one window, Hyperliquid perps in another; cross-venue generality is asserted only where a finding crosses systems (chase-state null, text–control contrast). **Model mix concentration.** DX Terminal Pro is one frozen model; DXAP is 86/91 qwen3.7-plus among active agents on Jul 22. **Probe scope.** H2 measures a proxy encoder (Qwen3.5-4B), not the production model’s activations. **Census drift.** DXAP all-history counts range 336–599 across vintages by inclusion rule; we quote 500–599 and use per-analysis denominators throughout. **Operator-reported figures.** The 99.9% settlement-success figure for DX Terminal Pro is single-source internal, with a policy-valid denominator.

12 Conclusion

Across six months and two production systems, the behavior of capital-managing LLM agents was set less by model reasoning than by the operating layer around it: a slider fixed leverage while volatility varied $5.7\times$; a render boundary moved selection $1.75\times$; one posture-slider cell held 62% of liquidations; a two-line bracket beat every discretionary exit policy we measured. The fleets produced no directional edge, and this record shows where the money goes instead: volatility-blind sizing, render-driven chase, and uncaptured excursion. For builders, the interventions are mechanical and cheap: bracket at the order path, scale leverage to realized volatility at the tool, A/B the render. For researchers, the canon of Section 9 is the part of this paper we expect to outlast the fleets themselves. For the lab, the model league closes one door (decision quality across frontier models is indistinguishable at this horizon), and the development program of Section 8 follows the evidence’s own ordering out of it: finish the measured operating-layer fixes, deploy the post-order risk check and the decision-provenance ledger, train on the environment the record itself provides, and let versioned benchmarks decide whether any of it worked.

Data availability

Aggregate statistics supporting every headline claim appear in the text and figures. Public datasets, dashboards, and per-claim evidence artifacts are published at <https://dxrg.ai/research>; the DX Terminal Pro public dashboard is at <https://dune.com/dxrg/dx-terminal-pro>. Raw DXAP production telemetry contains user identifiers and is available only as the aggregate extracts referenced above.

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